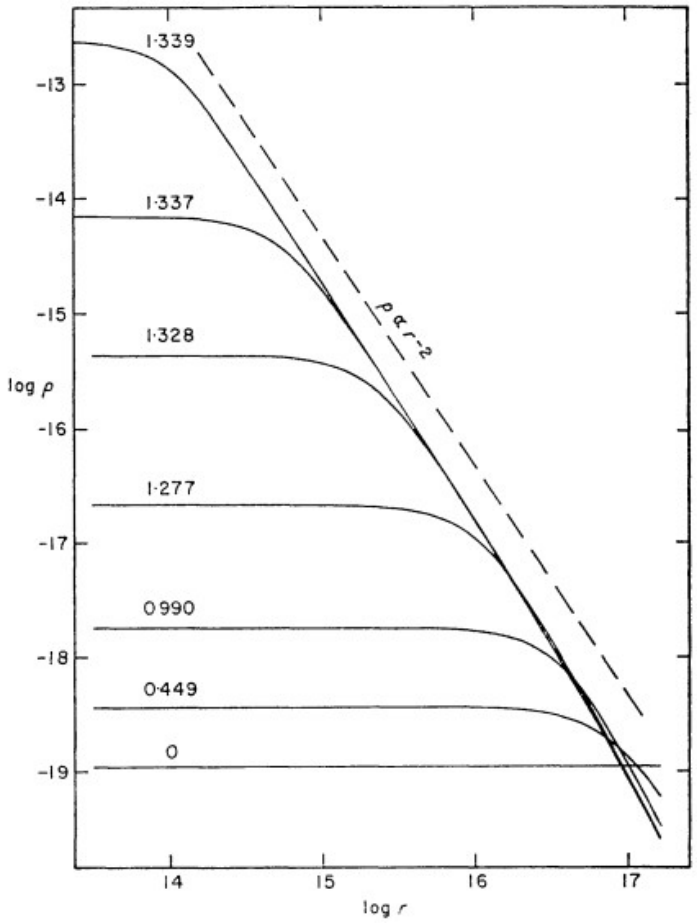
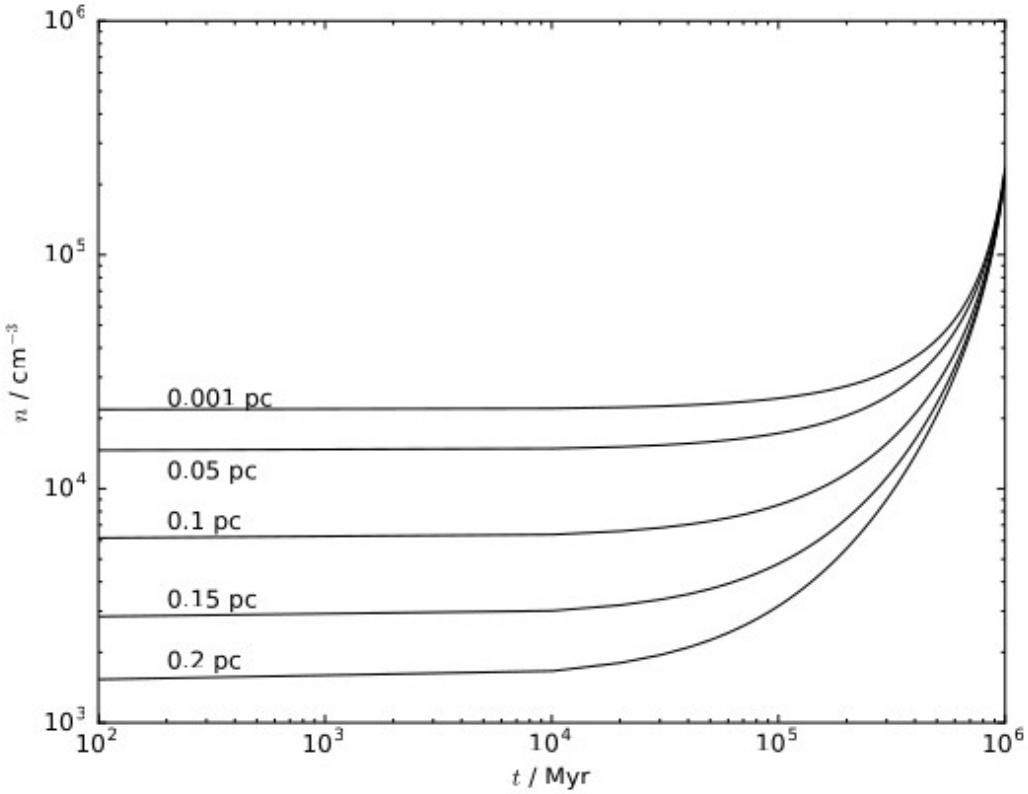
Collapse and post-processing modules in UCLCHEM

Felix Priestley

Cardiff University

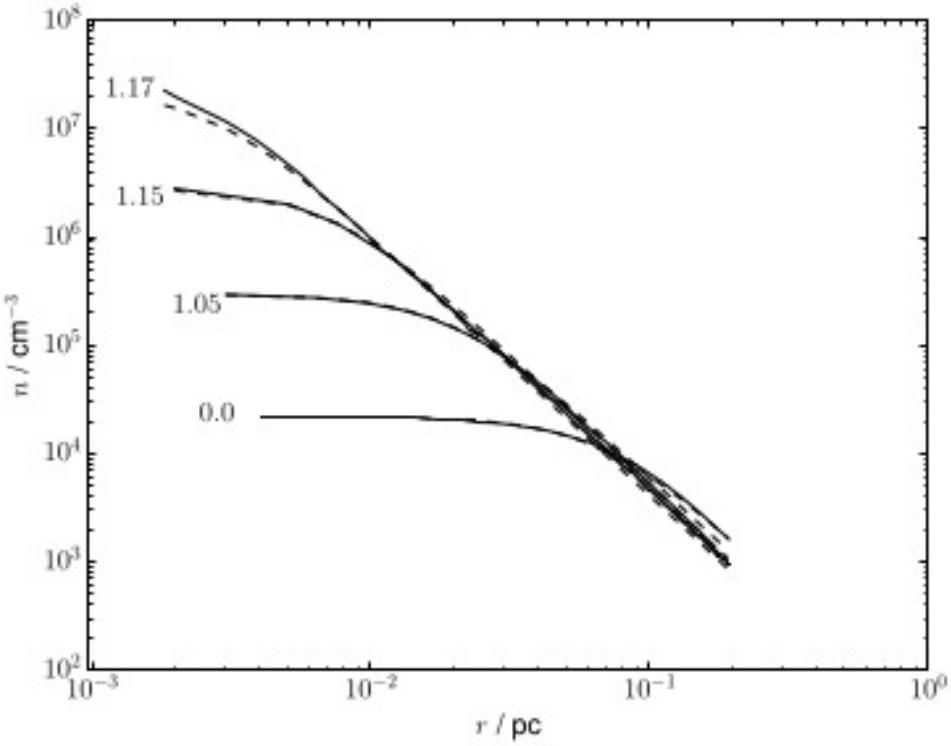
Collapse and post-processing modules in UCLCHEM





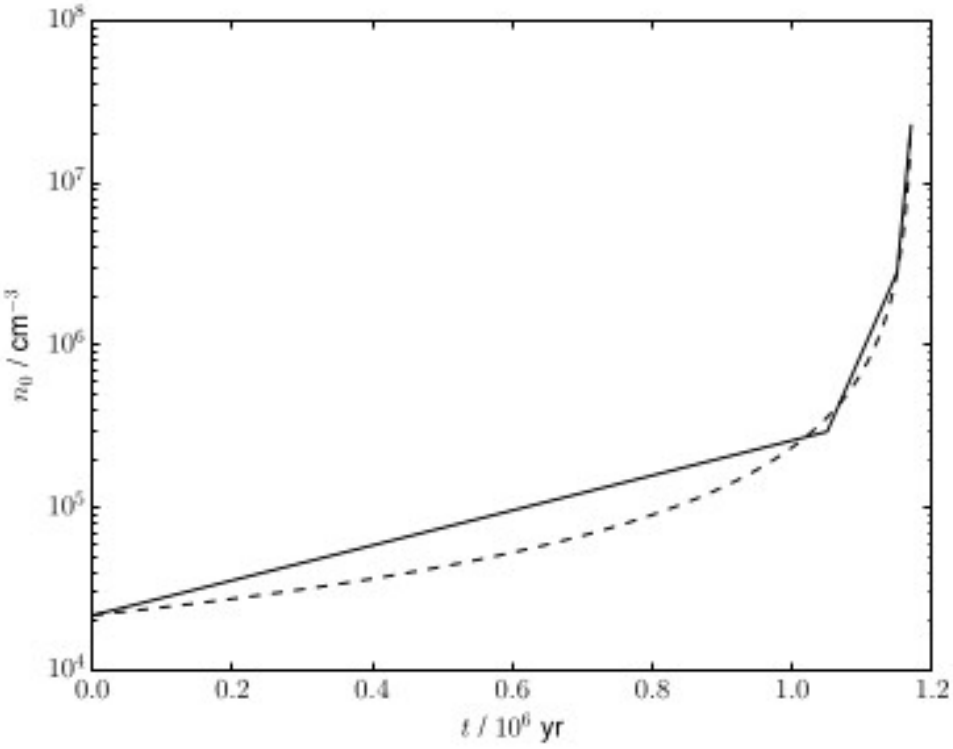
Larson (1969); Priestley et al. (2018)

Collapse and post-processing modules in UCLCHEM



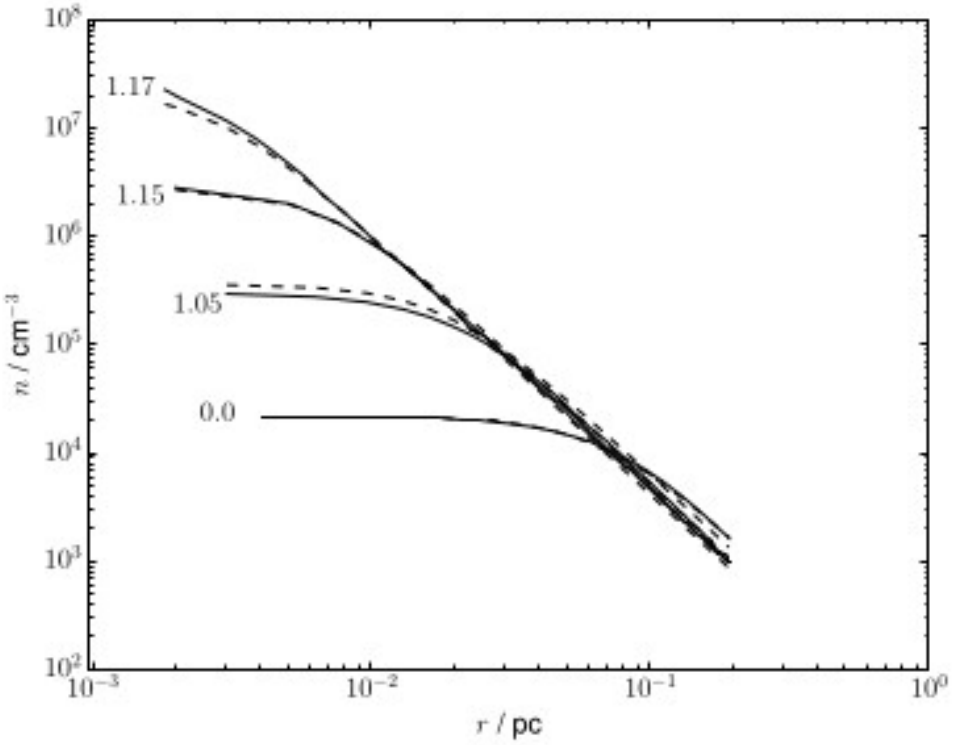
$$n(r) = \frac{n_0}{1 + (\frac{r}{r_0})^a}$$

Collapse and post-processing modules in UCLCHEM



$$\log n_0(t) = A(t_0 - t)^a + B$$

Collapse and post-processing modules in UCLCHEM



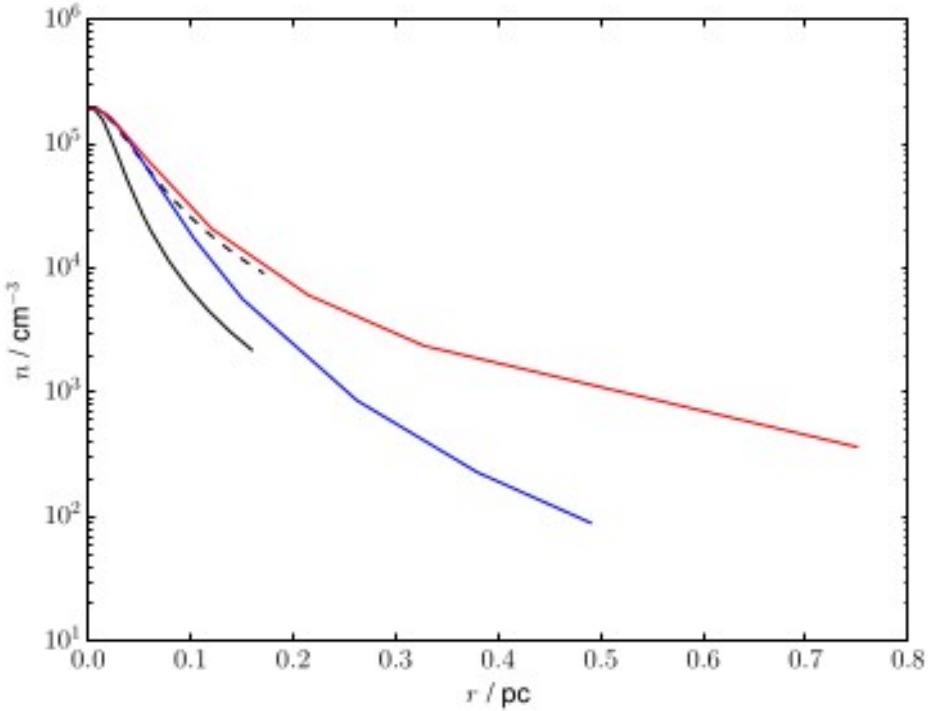
$$\log_{10} n_0(t) = 61.8\left(1.175 \times 10^6 - t\right)^{-0.01} - 49.4$$

$$\log_{10} r_0(t) = -28.5\left(1.175 \times 10^6 - t\right)^{-0.01} + 28.93$$

$$a(t) = 2.4$$

$$n(r) = \frac{n_0}{1 + \left(\frac{r}{r_0}\right)^a}$$

Collapse and post-processing modules in UCLCHEM



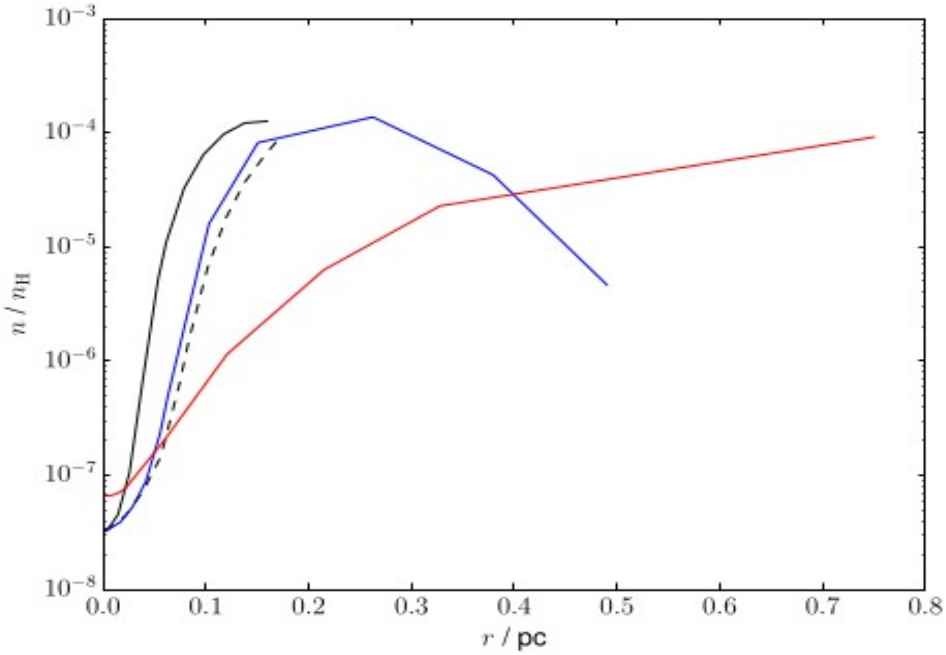
1: ‘normal’ Bonnor-Ebert sphere (black)

2: dense Bonnor-Ebert sphere (dashed)

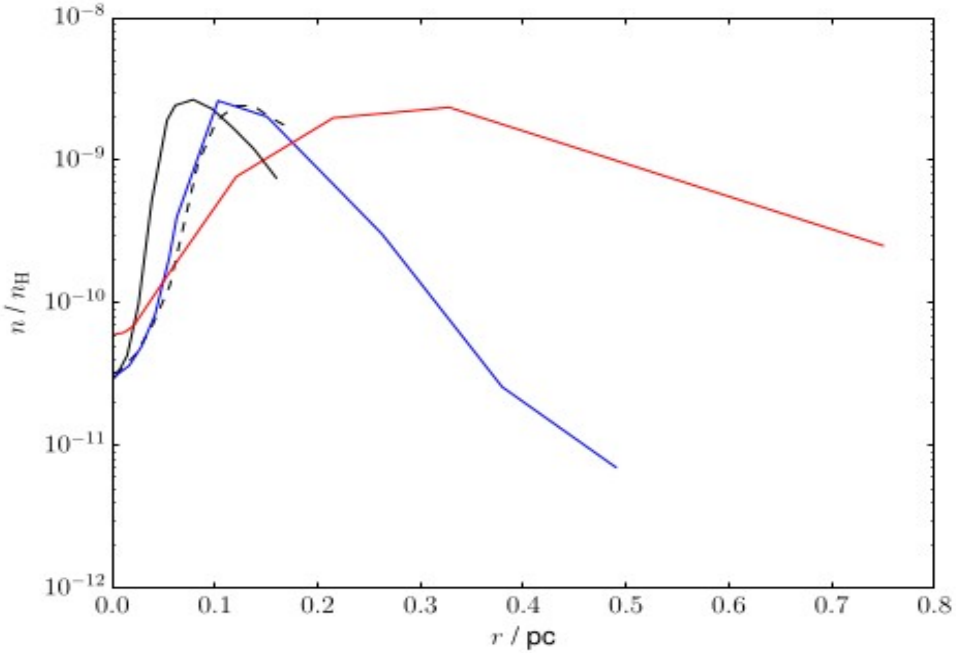
3: weakly-magnetised filament (blue)

4: strongly-magnetised core (red)

Collapse and post-processing modules in UCLCHEM



(a) CO



(c) HCO^+

Collapse and post-processing modules in UCLCHEM

```
SELECT CASE(collapse_mode)
   CASE(1)
       maxTime=1.175d6
       finalTime=0.97*maxTime
   CASE(2)
       maxTime=1.855d5
       finalTime=0.97*maxTime
   CASE(3)
   CASE(4)
   CASE DEFAULT
       write(*,*) "unacceptable collapse mode"
       successFlag=-1
       RETURN
END SELECT
```

collapse.f90

Collapse and post-processing modules in UCLCHEM

```
DO dstep=1,points
      parcelRadius(dstep)=dstep*rout/float(points)
END DO

IF (writePhysics) OPEN(unit=66,file=collapseFile,status='unknown',err=99)
IF (successFlag .lt. 0) THEN
   99 write(*,*) "could not open physics output file",collapseFile
   successFlag=-1
   RETURN
END IF
density=rhofit(rin,rho0fit(timeInYears),r0fit(timeInYears),afit(timeInYears))
IF (collapse_mode .le. 2) CALL findmassInRadius
```

collapse.f90

Collapse and post-processing modules in UCLCHEM

```
IF ((collapse_mode .le. 2)) THEN
    !I changed rin to rout
    CALL findNewRadius(massInRadius(dstep),rout,rho0fit(timeInYears),&
        &r0fit(timeInYears),afit(timeInYears),parcelRadius(dstep))
ELSE
    dt = targetTime - currentTime
    drad = vrfit(parcelRadius(dstep),rminfit(timeInYears),vminfit(timeInYears),avfit(timeInYears))*dt/pc
    parcelRadius(dstep) = parcelRadius(dstep) + drad
    IF (writePhysics) THEN
       write(66,*) timeInYears,parcelRadius(dstep),rhofit(parcelRadius(dstep),&
                &rho0fit(timeInYears),r0fit(timeInYears),afit(timeInYears)),&
                &vrfit(parcelRadius(dstep),rminfit(timeInYears),&
                &vminfit(timeInYears),avfit(timeInYears))
    END IF
END IF
density(dstep)=rhofit(parcelRadius(dstep),rho0fit(timeInYears),r0fit(timeInYears),afit(timeInYears))
```

collapse.f90

Collapse and post-processing modules in UCLCHEM

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Ambipolar diffusion and the molecular abundances in pre-stellar cores

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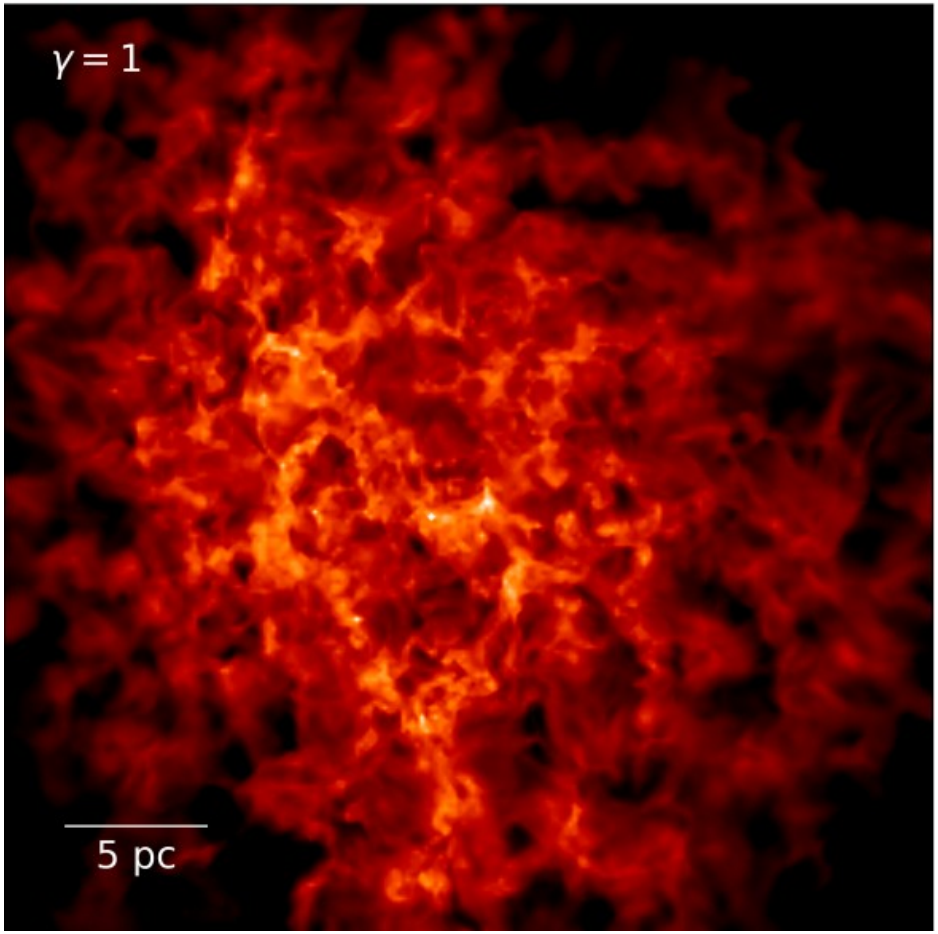
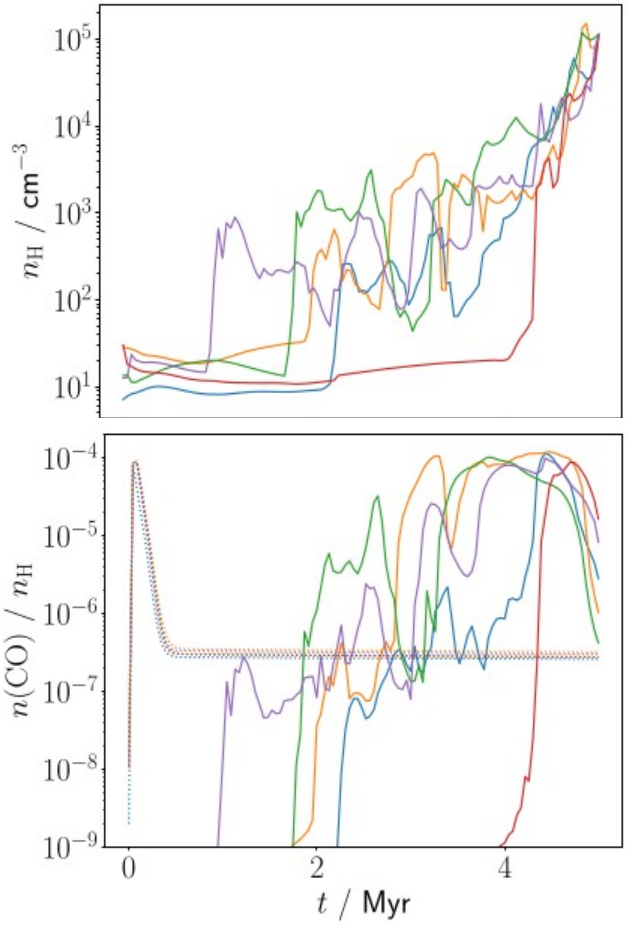
Collapse and post-processing modules in UCLCHEM





Priestley et al. (2023)

Collapse and post-processing modules in UCLCHEM

```
SUBROUTINE initializePhysics(successFlag, timegrid, densgrid, radgrid, zetagrid, gtempgrid,&
    &dtempgrid, usecoldens, timepoints, nhgrid, nh2grid, ncogrid, ncgrid)
  INTEGER, INTENT(OUT) :: successFlag
  INTEGER, INTENT(IN) :: timepoints
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: timegrid
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: densgrid
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: radgrid
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: zetagrid
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: gtempgrid
  DOUBLE PRECISION, INTENT(IN), DIMENSION(timePoints) :: dtempgrid
  LOGICAL, INTENT(IN) :: usecoldens
  DOUBLE PRECISION, INTENT(IN), OPTIONAL, DIMENSION(timePoints) :: nhgrid
  DOUBLE PRECISION, INTENT(IN), OPTIONAL, DIMENSION(timePoints) :: nh2grid
  DOUBLE PRECISION, INTENT(IN), OPTIONAL, DIMENSION(timePoints) :: ncogrid
  DOUBLE PRECISION, INTENT(IN), OPTIONAL, DIMENSION(timePoints) :: ncgrid
  ! write(*,*) 'Initialising postprocessing module'
```

postprocess.f90

Collapse and post-processing modules in UCLCHEM

```
! Store the parameter grids into a local module contexts
ltime(:) = timegrid
ldens(:) = densgrid
lradfield(:) = radgrid
lzeta(:) = zetagrid
lgtemp(:) = gtempgrid
ldtemp(:) = dtempgrid
! If we have custom column densities, allocate and store them
if (lusecoldens) then
! Only allocate the column densities if we need them:
  if (.not. allocated(lnh)) then
    allocate (lnh(timepoints), lnh2(timepoints), lnco(timepoints), lnc(timepoints))
  end if
  lnh(:) = nhgrid
  lnh2(:) = nh2grid
  lnco(:) = ncogrid
  lnc(:) = ncgrid
end if
```

ostprocess.f90

Collapse and post-processing modules in UCLCHEM

```
! Initialise values to t=1 and overwrite them.
tstep = 1
targettime = ltime(tstep)
density(dstep) = ldens(tstep)
gastemp(dstep) = lgtemp(tstep)
dusttemp(dstep) = ldtemp(tstep)
radfield = lradfield(tstep)
zeta = lzeta(tstep)
if (lusecoldens) then
  coldens(dstep) = lnh(tstep)
  av(dstep) = 5.348e-22 * coldens(dstep)
end if
! Set final time to end of tracer histories
finaltime = timegrid(timepoints)/seconds_per_year
```

postprocess.f90

Collapse and post-processing modules in UCLCHEM

```
SUBROUTINE updateTargetTime
  ! Set target time from postprocessing data
  targettime = ltime(tstep) + 1.*seconds_per_year

  ! write(*,"('Integrating chemistry to timestep ',I3

END SUBROUTINE updateTargetTime
```

postprocess.f90

Collapse and post-processing modules in UCLCHEM

```
.....................................................................
SUBROUTINE updatePhysics

  ! Update physical properties to values at tstep (== targettime)
  targettime = ltime(tstep)
  density(dstep) = ldens(tstep)
  gastemp(dstep) = lgtemp(tstep)
  dusttemp(dstep) = ldtemp(tstep)
  radfield = lradfield(tstep)
  zeta = lzeta(tstep)

  if (lusecoldens) then
    coldens(dstep) = lnh(tstep)
    av(dstep) = 5.348e-22 * coldens(dstep)
  end if
  ! If this is the last point, update tstep to move onto next time
  tstep = tstep + 1

END SUBROUTINE updatePhysics
```

postprocess.f90

Collapse and post-processing modules in UCLCHEM

fpriestley / neath Public

Code | Issues | Pull requests | Actions | Projects | Security | Insights

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fpriestley Update README.md 8f6e426 · 11 months ago 16 Commits

data	Add files via upload	11 months ago
output	Add files via upload	11 months ago
src	Add files via upload	11 months ago
LICENSE	Initial commit	3 years ago
README.md	Update README.md	11 months ago

README | MIT license

NEATH

NEATH (Non-Equilibrium Abundances Treated Holistically) is a framework for post-processing hydrodynamical simulations with a time-dependent astrochemical network, allowing the abundances of hundreds of molecules to be calculated at low computational cost. The code in this repository is an implementation based on a

About

Astrochemical model for hydrodynamical simulations

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No releases published

Packages

No packages published

Contributors 1

github.com/fpriestley/neath

Collapse and post-processing modules in UCLCHEM

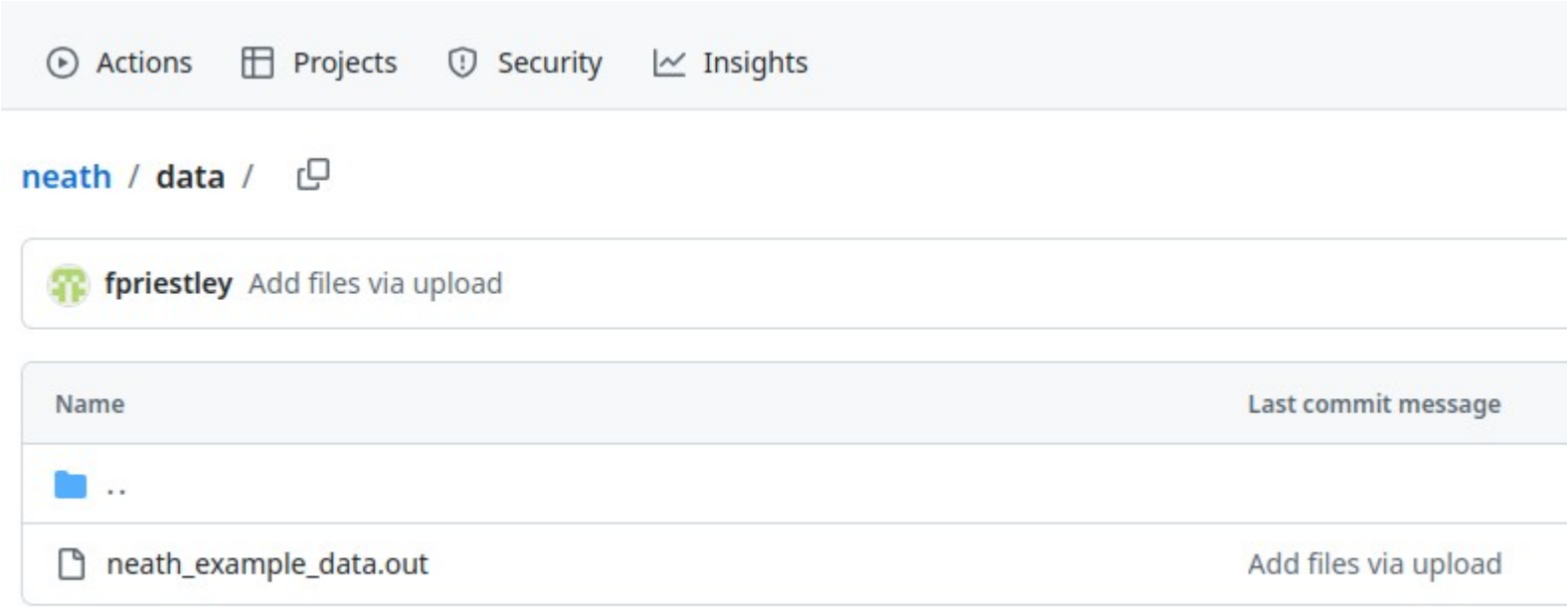
neath / src /

fpriestley Add files via upload

Name	Last commit message
..	
Makefile	Create Makefile
chem.f90	Add files via upload
defaultparameters.f90	Add files via upload
dvode.f90	Add files via upload
main.f90	Add files via upload
network.f90	Add files via upload
odes.f90	Add files via upload
postprocess.f90	Add files via upload
rates.f90	Add files via upload

github.com/fpriestley/neath

Collapse and post-processing modules in UCLCHEM



github.com/fpriestley/neath

Collapse and post-processing modules in UCLCHEM

raw.githubusercontent.com/fpriestley/neath/refs/heads/main/data/neath_example_data.out

0.000E+00	1.754E+19	-3.439E+19	-1.436E+19	1.332E+01	-7.407E+05	6.455E+04	1.210E+05	2.722E+02	0.000E+00	1.400E-04	1.000E-02	0.000E+00	0.000E+00	0.000E+00	1.500E+01
1.377E+12	1.666E+19	-3.423E+19	-1.424E+19	1.293E+01	-7.348E+05	6.238E+04	1.196E+05	1.890E+02	2.869E-04	1.154E-13	3.311E-03	4.489E+20	6.278E+16	3.794E+07	1.473E+01
2.753E+12	1.574E+19	-3.409E+19	-1.416E+19	1.204E+01	-7.352E+05	6.275E+04	1.181E+05	1.633E+02	5.754E-04	2.888E-13	1.777E-03	4.577E+20	1.385E+17	7.958E+07	1.473E+01
4.130E+12	1.481E+19	-3.393E+19	-1.409E+19	1.181E+01	-7.357E+05	6.375E+04	1.166E+05	1.673E+02	8.319E-04	5.885E-13	1.310E-03	4.630E+20	2.101E+17	1.564E+08	1.472E+01
5.506E+12	1.388E+19	-3.378E+19	-1.400E+19	1.156E+01	-7.355E+05	6.500E+04	1.164E+05	1.673E+02	1.071E-03	9.168E-13	1.112E-03	4.666E+20	2.708E+17	2.387E+08	1.471E+01
6.883E+12	1.288E+19	-3.368E+19	-1.385E+19	1.151E+01	-7.354E+05	6.586E+04	1.155E+05	1.642E+02	1.316E-03	1.336E-12	9.885E-04	4.748E+20	3.411E+17	3.521E+08	1.470E+01
8.259E+12	1.188E+19	-3.359E+19	-1.370E+19	1.149E+01	-7.354E+05	6.644E+04	1.129E+05	1.609E+02	1.560E-03	1.831E-12	9.161E-04	4.839E+20	4.119E+17	4.849E+08	1.469E+01
9.636E+12	1.087E+19	-3.349E+19	-1.355E+19	1.146E+01	-7.354E+05	6.675E+04	1.084E+05	1.584E+02	1.800E-03	2.387E-12	8.740E-04	4.912E+20	4.762E+17	6.250E+08	1.468E+01
1.101E+13	9.864E+18	-3.340E+19	-1.341E+19	1.139E+01	-7.352E+05	6.681E+04	1.027E+05	1.564E+02	2.038E-03	2.996E-12	8.489E-04	4.976E+20	5.481E+17	8.053E+08	1.467E+01
1.239E+13	8.856E+18	-3.330E+19	-1.327E+19	1.129E+01	-7.350E+05	6.667E+04	9.652E+04	1.551E+02	2.272E-03	3.728E-12	8.344E-04	5.093E+20	6.506E+17	1.096E+09	1.465E+01
1.377E+13	7.848E+18	-3.321E+19	-1.315E+19	1.115E+01	-7.347E+05	6.627E+04	9.048E+04	1.544E+02	2.503E-03	4.488E-12	8.273E-04	5.180E+20	7.192E+17	1.330E+09	1.464E+01
1.514E+13	6.840E+18	-3.312E+19	-1.303E+19	1.098E+01	-7.343E+05	6.558E+04	8.516E+04	1.541E+02	2.731E-03	5.298E-12	8.254E-04	5.267E+20	8.122E+17	1.657E+09	1.463E+01
1.652E+13	5.831E+18	-3.303E+19	-1.292E+19	1.078E+01	-7.340E+05	6.459E+04	8.090E+04	1.542E+02	2.954E-03	6.322E-12	8.273E-04	5.426E+20	9.098E+17	2.060E+09	1.461E+01
1.789E+13	4.822E+18	-3.294E+19	-1.281E+19	1.059E+01	-7.336E+05	6.330E+04	7.783E+04	1.545E+02	3.174E-03	7.534E-12	8.316E-04	5.616E+20	1.016E+18	2.597E+09	1.458E+01
1.927E+13	3.813E+18	-3.285E+19	-1.270E+19	1.041E+01	-7.333E+05	6.175E+04	7.600E+04	1.550E+02	3.390E-03	8.817E-12	8.374E-04	5.773E+20	1.117E+18	3.079E+09	1.456E+01
2.065E+13	2.804E+18	-3.277E+19	-1.260E+19	1.025E+01	-7.331E+05	6.001E+04	7.529E+04	1.553E+02	3.604E-03	1.037E-11	8.434E-04	5.960E+20	1.265E+18	4.356E+09	1.453E+01
2.202E+13	1.795E+18	-3.269E+19	-1.249E+19	1.013E+01	-7.330E+05	5.813E+04	7.555E+04	1.552E+02	3.816E-03	1.232E-11	8.483E-04	6.189E+20	1.453E+18	5.635E+09	1.449E+01
2.340E+13	7.859E+17	-3.261E+19	-1.239E+19	1.006E+01	-7.330E+05	5.618E+04	7.660E+04	1.545E+02	4.029E-03	1.527E-11	8.508E-04	6.550E+20	1.714E+18	7.523E+09	1.444E+01
2.478E+13	-2.235E+17	-3.253E+19	-1.228E+19	9.990E+00	-7.340E+05	5.505E+04	7.845E+04	1.524E+02	4.243E-03	1.912E-11	8.498E-04	6.986E+20	2.252E+18	1.153E+10	1.438E+01
2.615E+13	-1.233E+18	-3.246E+19	-1.217E+19	1.053E+01	-7.262E+05	5.055E+04	8.405E+04	1.612E+02	4.403E-03	3.156E-11	8.468E-04	7.691E+20	3.314E+18	2.266E+10	1.429E+01
2.744E+13	-1.953E+18	-3.240E+19	-1.191E+19	3.189E+01	-3.729E+05	2.481E+04	1.093E+05	3.563E+02	1.515E-02	7.592E-09	4.694E-04	8.627E+20	5.644E+18	1.041E+11	1.418E+01
2.882E+13	-2.284E+18	-3.231E+19	-1.171E+19	9.570E+01	-2.476E+05	7.927E+03	1.201E+05	6.211E+01	2.324E-02	3.389E-09	1.924E-04	9.463E+20	8.068E+18	2.736E+11	1.407E+01
3.019E+13	-2.577E+18	-3.230E+19	-1.155E+19	2.220E+02	-2.538E+05	-2.369E+03	1.194E+05	4.163E+01	2.427E-02	9.094E-10	1.385E-04	1.050E+21	1.080E+19	4.092E+11	1.393E+01
3.157E+13	-2.861E+18	-3.203E+19	-1.132E+19	1.614E+02	-1.718E+04	2.696E+04	9.902E+04	4.290E+01	2.430E-02	9.376E-10	1.465E-04	1.055E+21	1.128E+19	4.428E+11	1.393E+01
3.295E+13	-2.741E+18	-3.188E+19	-1.122E+19	2.918E+02	-1.873E+04	2.846E+04	1.025E+05	3.658E+01	3.878E-02	1.263E-09	1.197E-04	1.132E+21	1.529E+19	7.476E+11	1.381E+01
3.432E+13	-2.773E+18	-3.180E+19	-1.111E+19	2.579E+02	2.402E+04	2.826E+04	9.785E+04	3.514E+01	3.976E-02	9.935E-10	1.228E-04	1.147E+21	1.571E+19	6.715E+11	1.379E+01
3.570E+13	-2.859E+18	-3.164E+19	-1.112E+19	1.139E+02	2.064E+05	8.288E+04	4.712E+04	6.010E+01	2.736E-02	7.644E-09	1.662E-04	1.149E+21	1.671E+19	6.188E+11	1.381E+01
3.707E+13	-2.651E+18	-3.148E+19	-1.105E+19	1.786E+02	1.400E+05	8.241E+04	8.616E+04	4.032E+01	3.858E-02	2.936E-09	1.359E-04	1.185E+21	1.878E+19	1.212E+12	1.374E+01
3.845E+13	-2.493E+18	-3.138E+19	-1.091E+19	2.336E+02	9.887E+04	7.267E+04	1.194E+05	3.498E+01	4.548E-02	1.696E-09	1.215E-04	1.198E+21	2.281E+19	1.669E+12	1.369E+01
3.983E+13	-2.378E+18	-3.129E+19	-1.072E+19	2.807E+02	6.659E+04	5.791E+04	1.441E+05	3.165E+01	5.171E-02	1.368E-09	1.117E-04	1.231E+21	2.689E+19	1.795E+12	1.364E+01
4.120E+13	-2.328E+18	-3.119E+19	-1.050E+19	2.765E+02	6.562E+04	5.835E+04	1.569E+05	3.168E+01	5.327E-02	1.418E-09	1.132E-04	1.206E+21	2.525E+19	1.874E+12	1.366E+01
4.258E+13	-2.258E+18	-3.109E+19	-1.027E+19	2.067E+02	6.678E+04	6.209E+04	1.632E+05	3.638E+01	5.251E-02	2.799E-09	1.262E-04	1.202E+21	2.594E+19	2.311E+12	1.367E+01
4.396E+13	-2.201E+18	-3.098E+19	-1.008E+19	2.122E+02	6.082E+04	5.416E+04	1.663E+05	3.445E+01	5.486E-02	2.162E-09	1.243E-04	1.205E+21	2.840E+19	2.424E+12	1.366E+01

github.com/fpriestley/neath

Collapse and post-processing modules in UCLCHEM

neath / src / postprocess.f90

Code Blame 104 lines (82 loc) · 4.49 KB

```fortran

  CONTAINS
  !THIS IS WHERE THE REQUIRED PHYSICS ELEMENTS BEGIN.
  !YOU CAN CHANGE THEM TO REFLECT YOUR PHYSICS BUT THEY MUST BE NAMED ACCORDINGLY.

      SUBROUTINE initializePhysics
          IF (ALLOCATED(av)) DEALLOCATE(av,coldens,temp,dens)
          ALLOCATE(av(points),coldens(points),temp(points),dens(points))
          cloudSize=(rout-rin)*pc
          dens=initialDens
          temp=initialTemp

        do ii=1,points
           read(66,*) timegrid(1),(junk,kk=1,3),densgrid(ii,1),(junk,kk=1,3),tempgrid(ii,1),(junk,kk=1,3),avgrid(ii,1),nh2grid(ii,1),ncogrid(ii,1)
!           dens(ii) = densgrid(ii,1)/(1.4*mh)
           do jj=2,ntime
              read(66,*) timegrid(jj),(junk,kk=1,3),densgrid(ii,jj),(junk,kk=1,3),tempgrid(ii,jj),(junk,kk=1,3),avgrid(ii,jj),nh2grid(ii,jj),ncogrid(ii,jj)
           end do
        end do

        coldens = 2e4*sqrt(dens/(1.4*mh*G_N))
        av = baseAv + coldens*6e-22

      END SUBROUTINE
```

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Collapse and post-processing modules in UCLCHEM

```
!A full output of abundances is written by default. Additionally, name species here for
!a columnated output of time,density,temperature and abundances of those species
!Fortran will reject this array if species with shorter names are not padded with spaces at the end.
!array commented so it does not override array in input file.
!If no array is passed in input, no column file is written
outSpecies=(/'H2   ','E-   ','CO   ','NH3  ','HCN  ','N2H+ ','HCO+ ','CS   ','HNC  '/)

!writeStep sets how often columns written out. Columns written every n steps for writeStep=n.
writeStep=1

!If readAbund=1, starting abundances are read from abundFile
!If readAbund=0, final abundances are written to abundFile
abundFile="output/startcloud.dat"
!Full output written to outputFlie
outputFile='output/'//TRIM(trajecfile)
!columnated output of time,dens,temp and outSpecies written to column file
columnFile='output/columncloud.dat'
```

github.com/fpriestley/neath

Collapse and post-processing modules in UCLCHEM

neath / output /

fpriestley Add files via upload

Name	Last commit message
..	
neath_example_data.out	Add files via upload

github.com/fpriestley/neath

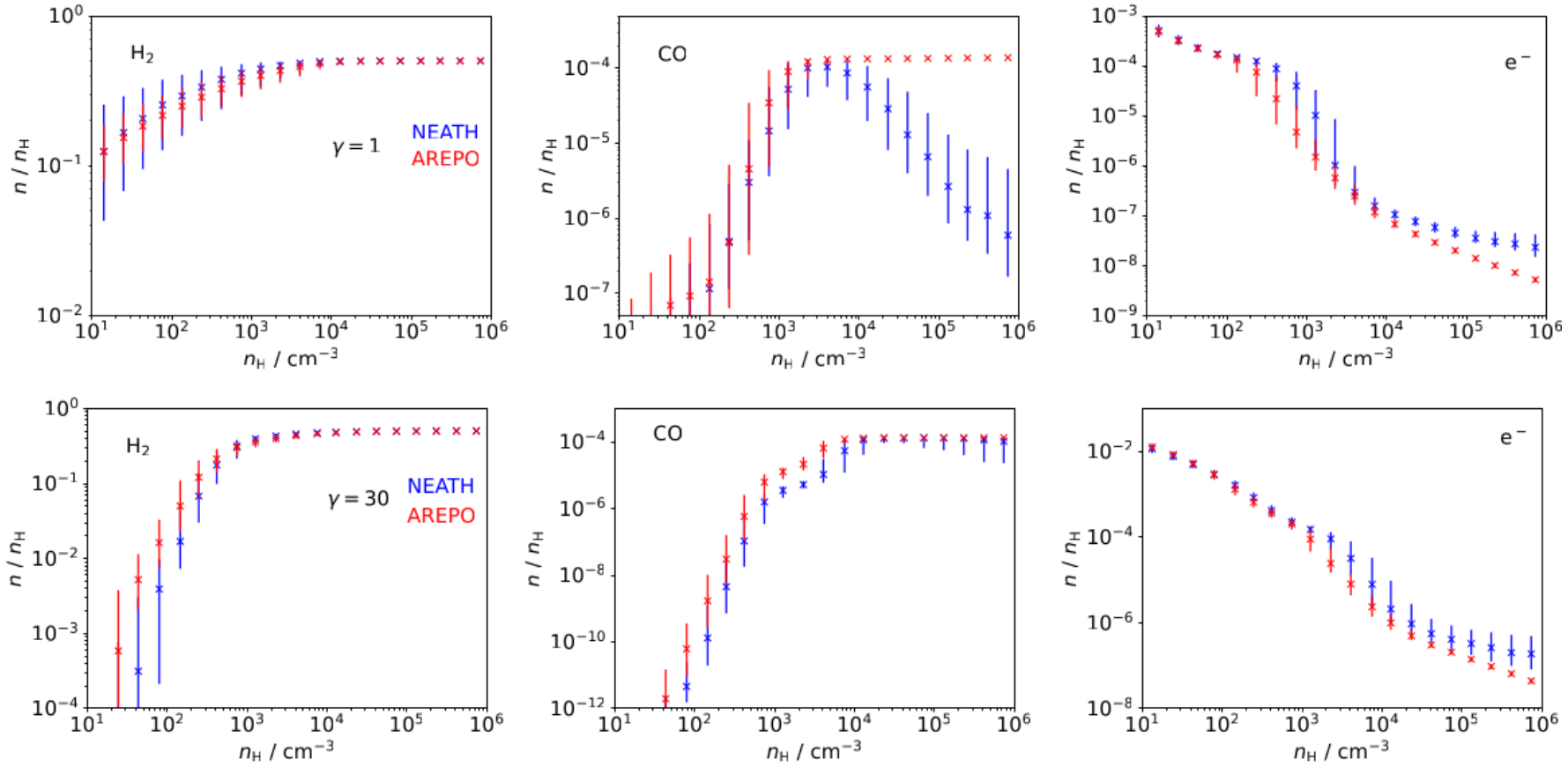
Collapse and post-processing modules in UCLCHEM

raw.githubusercontent.com/fpriestley/neath/refs/heads/main/output/neath_example_data.out

1.000E+00	1.332E+01	2.722E+02	0.000E+00	1.595E-08	1.407E-04	9.740E-18	1.000E-30	9.482E-28	1.000E-30	3.707E-24	1.000E-30	1.000E-30
1.000E+00	7.535E+00	2.722E+02	0.000E+00	9.064E-09	1.407E-04	3.293E-18	6.513E-13	1.207E-20	7.191E-24	4.087E-17	9.822E-23	7.405E-21
1.000E+00	5.132E+00	2.722E+02	0.000E+00	6.205E-09	1.407E-04	1.422E-18	4.436E-13	5.008E-21	3.339E-24	1.898E-17	3.110E-23	3.449E-21
1.000E+00	2.428E+01	2.722E+02	0.000E+00	2.899E-08	1.407E-04	4.855E-17	2.099E-12	2.166E-19	7.407E-23	4.217E-16	3.234E-21	7.472E-20
1.000E+00	7.349E+00	2.722E+02	0.000E+00	8.843E-09	1.407E-04	3.116E-18	6.352E-13	1.139E-20	6.841E-24	3.888E-17	9.114E-23	7.046E-21
1.000E+00	1.291E+01	2.722E+02	0.000E+00	1.546E-08	1.407E-04	1.108E-17	1.116E-12	4.376E-20	2.105E-23	1.197E-16	4.914E-22	2.154E-20
1.000E+00	1.691E+01	2.722E+02	0.000E+00	2.022E-08	1.407E-04	2.069E-17	1.462E-12	8.567E-20	3.605E-23	2.051E-16	1.100E-21	3.670E-20
1.000E+00	1.714E+01	2.722E+02	0.000E+00	2.049E-08	1.407E-04	2.136E-17	1.481E-12	8.864E-20	3.704E-23	2.107E-16	1.145E-21	3.769E-20
1.000E+00	6.629E+00	2.722E+02	0.000E+00	7.986E-09	1.407E-04	2.483E-18	5.730E-13	8.970E-21	5.568E-24	3.165E-17	6.694E-23	5.740E-21
1.000E+00	6.724E+00	2.722E+02	0.000E+00	8.099E-09	1.407E-04	2.562E-18	5.812E-13	9.269E-21	5.729E-24	3.256E-17	6.985E-23	5.905E-21
1.000E+00	2.347E+01	2.722E+02	0.000E+00	2.802E-08	1.407E-04	4.479E-17	2.029E-12	1.983E-19	6.924E-23	3.941E-16	2.923E-21	6.991E-20
1.000E+00	1.153E+01	2.722E+02	0.000E+00	1.382E-08	1.407E-04	8.557E-18	9.966E-13	3.320E-20	1.681E-23	9.556E-17	3.506E-22	1.722E-20
1.000E+00	2.448E+01	2.722E+02	0.000E+00	2.922E-08	1.407E-04	4.951E-17	2.116E-12	2.213E-19	7.529E-23	4.286E-16	3.314E-21	7.593E-20
1.000E+00	7.137E+00	2.722E+02	0.000E+00	8.591E-09	1.407E-04	2.921E-18	6.169E-13	1.064E-20	6.453E-24	3.667E-17	8.350E-23	6.648E-21
1.000E+00	3.316E+01	2.722E+02	0.000E+00	3.955E-08	1.407E-04	1.024E-16	2.866E-12	4.932E-19	1.376E-22	7.838E-16	8.169E-21	1.373E-19
1.000E+00	2.737E+01	2.722E+02	0.000E+00	3.266E-08	1.407E-04	6.462E-17	2.366E-12	2.965E-19	9.398E-23	5.352E-16	4.619E-21	9.445E-20
1.000E+00	7.783E+00	2.722E+02	0.000E+00	9.359E-09	1.407E-04	3.538E-18	6.727E-13	1.302E-20	7.671E-24	4.360E-17	1.082E-22	7.897E-21
1.000E+00	4.525E+00	2.722E+02	0.000E+00	5.483E-09	1.407E-04	1.084E-18	3.911E-13	3.777E-21	2.597E-24	1.476E-17	2.133E-23	2.684E-21
1.000E+00	1.196E+01	2.722E+02	0.000E+00	1.433E-08	1.407E-04	9.302E-18	1.034E-12	3.630E-20	1.808E-23	1.028E-16	3.911E-22	1.851E-20
1.000E+00	6.063E+00	2.722E+02	0.000E+00	7.313E-09	1.407E-04	2.042E-18	5.240E-13	7.310E-21	4.659E-24	2.648E-17	5.124E-23	4.807E-21
1.000E+00	7.932E+00	2.722E+02	0.000E+00	9.537E-09	1.407E-04	3.690E-18	6.856E-13	1.361E-20	7.967E-24	4.529E-17	1.145E-22	8.200E-21
1.000E+00	2.654E+01	2.722E+02	0.000E+00	3.168E-08	1.407E-04	6.003E-17	2.294E-12	2.734E-19	8.840E-23	5.034E-16	4.214E-21	8.893E-20
1.000E+00	2.433E+01	2.722E+02	0.000E+00	2.905E-08	1.407E-04	4.879E-17	2.103E-12	2.178E-19	7.437E-23	4.234E-16	3.254E-21	7.502E-20
1.000E+00	1.425E+01	2.722E+02	0.000E+00	1.705E-08	1.407E-04	1.391E-17	1.232E-12	5.584E-20	2.564E-23	1.458E-16	6.601E-22	2.618E-20
1.000E+00	7.816E+00	2.722E+02	0.000E+00	9.399E-09	1.407E-04	3.571E-18	6.756E-13	1.315E-20	7.737E-24	4.397E-17	1.096E-22	7.964E-21
1.000E+00	6.694E+00	2.722E+02	0.000E+00	8.064E-09	1.407E-04	2.537E-18	5.786E-13	9.174E-21	5.680E-24	3.227E-17	6.892E-23	5.853E-21
1.000E+00	7.918E+00	2.722E+02	0.000E+00	9.520E-09	1.407E-04	3.676E-18	6.844E-13	1.355E-20	7.939E-24	4.513E-17	1.139E-22	8.171E-21
1.000E+00	1.332E+01	2.722E+02	0.000E+00	1.595E-08	1.407E-04	1.190E-17	1.151E-12	4.726E-20	2.241E-23	1.274E-16	5.396E-22	2.291E-20
1.000E+00	9.473E+00	2.722E+02	0.000E+00	1.137E-08	1.407E-04	5.486E-18	8.188E-13	2.070E-20	1.136E-23	6.455E-17	1.948E-22	1.166E-20
1.000E+00	9.631E+00	2.722E+02	0.000E+00	1.156E-08	1.407E-04	5.694E-18	8.324E-13	2.153E-20	1.174E-23	6.672E-17	2.047E-22	1.205E-20
1.000E+00	1.060E+01	2.722E+02	0.000E+00	1.271E-08	1.407E-04	7.070E-18	9.162E-13	2.709E-20	1.421E-23	8.079E-17	2.726E-22	1.458E-20
1.000E+00	8.256E+00	2.722E+02	0.000E+00	9.922E-09	1.407E-04	4.034E-18	7.136E-13	1.495E-20	8.630E-24	4.906E-17	1.291E-22	8.879E-21
1.000E+00	1.285E+01	2.722E+02	0.000E+00	1.539E-08	1.407E-04	1.096E-17	1.111E-12	4.326E-20	2.086E-23	1.186E-16	4.847E-22	2.134E-20
1.000E+00	1.522E+01	2.722E+02	0.000E+00	1.821E-08	1.407E-04	1.620E-17	1.315E-12	6.578E-20	2.923E-23	1.663E-16	8.035E-22	2.982E-20
1.000E+00	8.245E+00	2.722E+02	0.000E+00	9.909E-09	1.407E-04	4.022E-18	7.126E-13	1.490E-20	8.607E-24	4.893E-17	1.286E-22	8.855E-21
1.000E+00	1.086E+01	2.722E+02	0.000E+00	1.302E-08	1.407E-04	7.469E-18	9.386E-13	2.872E-20	1.491E-23	8.480E-17	2.931E-22	1.529E-20
1.000E+00	1.324E+01	2.722E+02	0.000E+00	1.585E-08	1.407E-04	1.174E-17	1.144E-12	4.656E-20	2.214E-23	1.259E-16	5.299E-22	2.264E-20
1.000E+00	6.065E+00	2.722E+02	0.000E+00	7.315E-09	1.407E-04	2.044E-18	5.242E-13	7.315E-21	4.662E-24	2.650E-17	5.129E-23	4.810E-21
1.000E+00	9.899E+00	2.722E+02	0.000E+00	1.188E-08	1.407E-04	6.057E-18	8.556E-13	2.299E-20	1.240E-23	7.048E-17	2.222E-22	1.273E-20

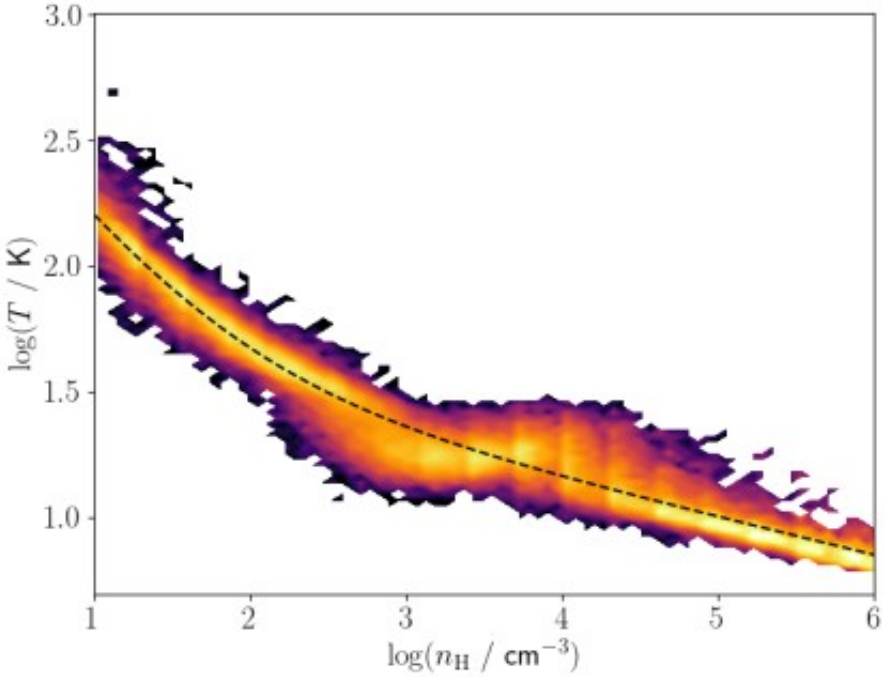
github.com/fpriestley/neath

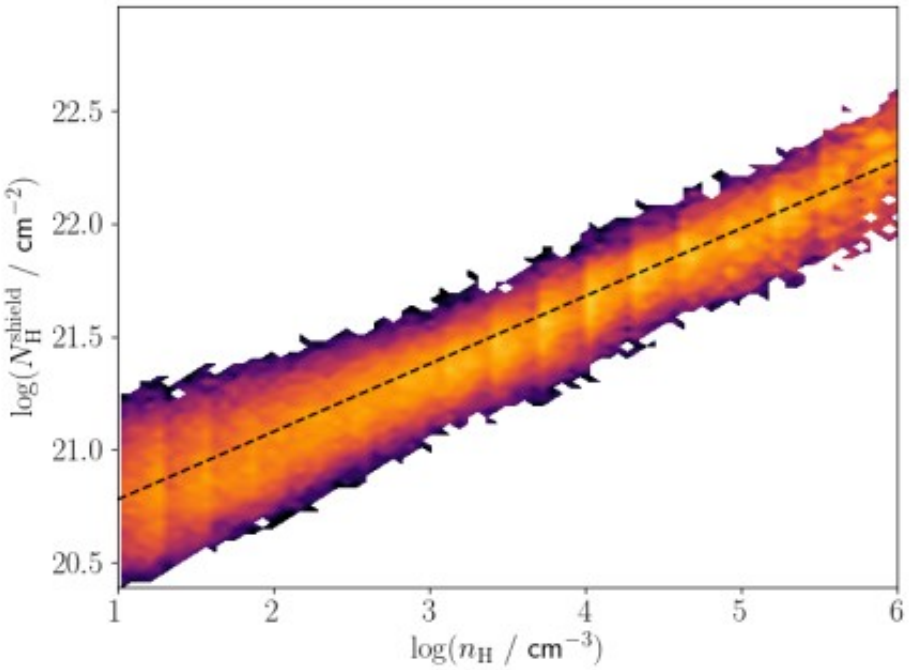
Collapse and post-processing modules in UCLCHEM



NEATH benchmarking (Priestley et al. 2023, 2026)

Collapse and post-processing modules in UCLCHEM





Density-temperature shielding relationships (Priestley et al. 2023)